

Ilyan EL HAJJAJI

Engineering Student in Electronics & Automation (M.Sc. / Bac+5) *Toulouse*
INP-ENSEEIHT
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Profile

Rigorous • Hands-on • Open-minded

First-year Master's degree engineering student (5-year Bac+5 programme) in the 3EA department. Driven by **semiconductors** and **control systems**. Seeking an international academic exchange and a technical internship.

Target Mobility

National Taiwan University
IC Design & semiconductor foundries

EPFL (Switzerland)
Microengineering & precision control

TU Munich (Germany)
Mechatronics & embedded systems

Applied Technical Skills

Control & Hardware
Closed-loop PID tuning, analog & digital circuitry, Hall-effect and infrared sensors (applied in lab prototypes).

Software & Modeling
MATLAB, Simulink (dynamic simulation), Python, C, OCaml, Git/GitHub, LaTeX.

Languages

- **French:** Native
- **English:** Fluent (B2+)
- **Spanish:** Intermediate (B2)
- **Mandarin Chinese:** A1/A2 (ongoing)
- **German / Arabic:** Beginner

Activities & Soft Skills

- **Club Football (4 years):** Developed team spirit, fast communication under pressure, and resilience in competition.
- **Civic & Sustainability Path:** Climate Fresk, Sulitest, and workplace inclusion workshop – active listening and ethical awareness.
- **International Travel:** Cross-cultural adaptability and open-mindedness.

Education

Toulouse INP - ENSEEIHT *2026 – Present*
M.Sc. in Electrical, Electronics & Automation Engineering (Bac+5 Programme)
Toulouse, France

- Coursework: Analog and digital electronics, continuous and discrete-time control systems, signal processing, electrical energy, and dynamic modeling.
- **Skills gained:** Circuit prototyping, frequency-domain analysis, collaborative project management, and technical communication.

Lycée Pierre Corneille *2023 – 2026*
CPGE MPI (Intensive Preparatory Classes in Maths, Physics & Computer Science) *Rouen, France*

- Three-year intensive preparation for the nationwide competitive entrance exams to French *Grandes Écoles*.
- **Skills gained:** Mathematical rigor, complex problem-solving under time pressure, data structures and algorithms (C, OCaml), and strong work ethic.

Sainte-Famille High School *2020 – 2023*
French Scientific Baccalaureate (Majors: Mathematics & Physics-Chemistry) – Highest Honors *Amiens, France*

Engineering Projects & Skills Demonstrated

Active Feedback Magnetic Levitation System

- Designed and soldered an analog closed-loop circuit to keep a steel ball suspended at a fixed point in the air without physical contact.
- Tuned a real-time PID controller using Hall-effect sensor feedback to dynamically reject external disturbances.
- **Skills gained:** Electromagnet sizing, bridging theoretical equations with real hardware noise, sensor calibration, and systematic lab troubleshooting.

Automotive ABS Braking Optimization Model

- Simulated wheel-road slip dynamics during emergency braking to prevent wheel lock-up and shorten stopping distances.
- Designed, tested, and benchmarked 3 control laws in Simulink: **relay control**, **classical PID**, and **pseudo-AI optimized PID**.
- **Skills gained:** Non-linear dynamic modeling in MATLAB/Simulink, critical benchmarking of control algorithms, and parameter optimization.

Autonomous Obstacle-Avoidance Mobile Robot Car

- Built an autonomous wheeled robot capable of navigating an indoor room while detecting and avoiding walls.
- **Skills gained:** Real-time reactive programming (C/Python), infrared sensor interfacing, motor control, and hardware-software integration.

Research & International Exposure

- **IEEE International Conference (Seoul, South Korea, 2023):** Attended technical sessions on next-generation semiconductors and automation. *Skills gained: scientific listening in English, technology watch, and motivation for an academic exchange in East Asia.*
- **Research Immersion at MIS Laboratory (Amiens, France):** Shadowed researchers working on non-linear control algorithms and mobile robotics testbeds. *Skills gained: understanding of academic R&D methodology and experimental validation.*